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Fast Payment Ecosystem Models and Performance in Brazil, the United States, and China

Prepared by Yibin Mu

WP/26/246

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Prepared by Yibin Mu

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ABSTRACT: This paper compares fast payment ecosystems in Brazil, the United States, and China, focusing on payment infrastructure, schemes, and applications. It examines how governance, interoperability, pricing, and access shape adoption, competition, trust, and resilience. The analysis finds that outcomes depend less on public versus private provision than on how these features interact under domestic conditions. Brazil illustrates rapid diffusion through common rules and low-friction access; the U.S. case shows that multiple infrastructures need not preclude interoperability; and China demonstrates how platform integration can generate scale while increasing concentration and oversight challenges. The paper provides a qualitative assessment of institutional tradeoffs rather than causal estimates or a ranking of models.

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WORKING PAPERS

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Prepared by Yibin Mu[1]

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Contents

Acronyms and Abbreviations **4**

Executive Summary **5**

I. Introduction **6**

II. Analytical Framework and Global Context **8**
A. Analytical Dimensions for Comparing Fast Payment System Models 8
B. Payment Systems, Payment Schemes, and Ecosystem Architecture 9
C. Global Patterns and Institutional Models 9
D. Policy Relevance for Emerging and Developing Economies 10
E. Why Brazil, the United States, and China 10

III. Country Case Studies **11**
A. Brazil: Pix as a Unified Public Rail Model 11
B. United States: A Decentralized Hybrid Multi-Rail Model 12
C. China: Centralized Back-End Infrastructure with a Platform-Led Retail Ecosystem 15

IV. Cross-Country Comparative Analysis **18**
A. Comparative Overview of the Three Models 18
B. Governance and Institutional Design 18
C. System Architecture and Interoperability 19
D. Pricing, Access, and Adoption Dynamics 19
E. Competition, Concentration, and Innovation 20
F. Risk, Oversight, and Institutional Tradeoffs 20
G. Comparative Synthesis 20

V. Economic and Financial Implications **22**
A. Financial Inclusion and Payment Access 22
B. Payment Efficiency, Transaction Costs, and Liquidity 22
C. Market Structure and Financial Sector Competition 23
D. Government Payments and Public Financial Management 24
E. Financial Stability, Risk, and Operational Resilience 24
F. Digital Transformation and Wider Economic Effects 25
G. Synthesis 25

VI. Comparative Considerations for System Design **26**
A. Governance, Interoperability, and Access 26
B. Adoption, User Friction, and Incentives 26
C. Competition and Fragmentation 27
D. Risk Management, Oversight, and Consumer Protection 27
E. Institutional Capacity and Country Context 28

VII. Conclusion **30**

References **31**

FIGURES

1. Analytical Framework for Comparing Fast Payment Systems 8
2. United States: A Decentralized Hybrid Multi-Rail Model 13
3. China: Centralized Back-End Infrastructure with a Platform-Led Retail Ecosystem 15

TABLE

1. Major Instant Payment Platforms: Comparative Analysis 21

Acronyms and Abbreviations

Acronym	Full term
ACH	Automated Clearing House
API	Application programming interface
AML/CFT	Anti-money laundering and combating the financing of terrorism
BIS	Bank for International Settlements
BCB	Banco Central do Brasil
CBDC	Central bank digital currency
CFPB	Consumer Financial Protection Bureau
CNAPS	China National Advanced Payment System
DICT	Directory of Transactional Account Identifiers
e-CNY	Digital yuan
FedNow	FedNow Service
FEBRABAN	Brazilian Federation of Banks
FGV	Fundação Getulio Vargas
FPE	Fast payment ecosystem
IBPS	Internet Banking Payment System
FSB	Financial Stability Board
IMF	International Monetary Fund
ISO 20022	International Organization for Standardization 20022 financial messaging standard
NUCC	NetsUnion Clearing Corporation
P2P	Peer-to-peer
PBoC	People’s Bank of China
QR	Quick response
RTGS	Real-time gross settlement
RTP	Real-Time Payments network
SPI	Instant Payment System
TED	Transferência Eletrônica Disponível
TCH	The Clearing House Payments Company
UPI	Unified Payments Interface

Executive Summary

Fast payment ecosystems (FPEs) have become core financial infrastructure, but countries have pursued different institutional configurations to achieve scale, usability, and trust. In this paper, an FPE encompasses both the payment system or settlement infrastructure and the schemes and applications through which fast payment services are delivered. The analysis is qualitative and comparative: it examines how three design dimensions—governance and ownership, architecture and interoperability, and pricing and access—shape adoption and usability, competition and concentration, and trust and resilience across Brazil, the United States, and China.

Three findings stand out. First, interoperability should not be inferred from the number of payment infrastructures alone. Pix is interoperable by design because participating providers operate under a common scheme with shared rules, standards, identifiers, and participation arrangements, supported by the SPI settlement infrastructure. The United States has multiple instant-payment infrastructures—FedNow and RTP—but their coexistence does not itself prevent interoperability; the more relevant constraint is incomplete alignment across schemes, providers, directories, standards, and participation arrangements. China combines centralized public infrastructure with dominant private wallet ecosystems, achieving broad functional usability but historically less complete interoperability across platforms.

Second, adoption reflects the interaction of institutional design and country conditions. Brazil's rapid diffusion was supported by mandatory participation by large institutions, zero-cost retail transfers, simple aliases, merchant usability, and coordinated network formation. China's scale reflected low-cost QR acceptance, a less-entrenched card infrastructure, platform integration, escrow-based trust in early e-commerce, commercial incentives, and a relatively permissive early regulatory environment. U.S. adoption has been more gradual because instant payments coexist with mature card and ACH systems, established rewards programs, decentralized participation, and heterogeneous provider incentives.

Third, the broader effects of fast payment ecosystems depend on how infrastructure, schemes, applications, and oversight arrangements work together. Where low-friction access, merchant acceptance, and routine use reinforce one another, FPEs can support payment efficiency, inclusion, government payment modernization, and wider digitalization. These gains may be accompanied by different risks: common public arrangements concentrate operational and fraud-management responsibilities; plural infrastructures complicate coordination; and platform-led ecosystems can increase concentration, data-governance, and supervisory challenges.

The comparison is intended as a structured assessment of institutional tradeoffs, not as a ranking of models, a set of causal estimates, or institutional policy advice. The relevance of each configuration depends on domestic institutional capacity, market structure, legacy payment arrangements, and policy objectives.

I. Introduction

Fast payment ecosystems (FPEs) have become core financial infrastructure in a growing number of economies. Following CPMI, a fast payment system (FPS) enables the transmission of payment messages and the availability of final funds to the payee in real or near-real time, on as close to a 24/7 basis as possible. The broader FPE examined in this paper encompasses both the payment system or settlement infrastructure and the payment scheme, including the rules, standards, participation arrangements, and user-facing applications through which fast payment services are provided. Not every transaction initiated through an FPE application necessarily settles through an FPS: some wallet-to-wallet transactions may be recorded on an internal ledger, while funding, withdrawal, or cross-provider transactions may use bank, card, ACH, or instant-payment infrastructures. This broader framing is particularly important for comparing Brazil, the United States, and China on a like-for-like basis.

Countries have pursued markedly different institutional pathways. Some combine a common public scheme with public settlement infrastructure; others rely on multiple public and private infrastructures and schemes; still others combine centralized public infrastructure with dominant private platforms. These differences matter because ecosystem outcomes depend not only on technology or ownership, but also on governance, interoperability, pricing, access, legacy payment arrangements, and the organization of the surrounding financial and digital ecosystem. The central analytical question is therefore how alternative institutional configurations shape scale, usability, competition, trust, and resilience under different domestic conditions.

The paper addresses this question through a structured comparison of Brazil, the United States, and China. The cases are selected purposively rather than as a representative sample. Brazil illustrates a unified public scheme supported by public settlement infrastructure; the United States represents a hybrid model in which public and private infrastructures coexist alongside multiple schemes and front-end applications; and China combines centralized public infrastructure with predominantly private platform ecosystems at the retail layer. India’s Unified Payments Interface is discussed as an additional comparator showing how an open public protocol can support competing consumer-facing applications on a common interoperable network.

A growing literature examines the design, adoption, interoperability, and financial-sector effects of fast and digital payment systems. Recent studies highlight the roles of interoperability, network effects, coordinated supply-side integration, adoption costs, competition, deposit funding, and bank liquidity (Argente et al., 2025; Copestake et al., 2025a, 2025b; Gonzalez, Ma, and Zeng, 2025; Kahn, 2024; Liang, Sampaio, and Sarkisyan, 2025; Sampaio and Ornelas, 2024; Sarkisyan, 2024; Xu, 2026). This paper complements that literature by comparing how these mechanisms operate across three contrasting institutional configurations. Its contribution is qualitative and comparative rather than econometric: it distinguishes design choices—governance and ownership, architecture and interoperability, and pricing and access—from outcomes, including adoption and usability, competition and concentration, and trust and resilience. The paper identifies recurring mechanisms and tradeoffs while drawing on existing empirical studies where causal evidence is available.

The scope of the paper is deliberately domestic and institutional. While cross-border payment linkages are an important and growing policy topic, this paper does not examine the cross-border dimension of fast payment systems in detail. Cross-border FPS integration raises additional issues—including foreign-exchange conversion, settlement finality across jurisdictions, AML/CFT compliance, sanctions screening, data governance, and legal and supervisory coordination—that go beyond the comparative domestic framework developed here.

The paper advances four comparative propositions. Institutional coherence matters more than ownership alone in explaining outcomes. System-level interoperability is the main mechanism through which network

effects scale. Platform-led adoption can substitute for formal interoperability up to a point, but with different governance risks. Plurality of rails may support resilience and innovation, while weakening unified retail diffusion when coordination mechanisms are limited.

The rest of the paper proceeds as follows. Section II sets out the analytical framework and global context for comparing FPE models. Section III presents the three country case studies. Section IV compares the cases across governance, interoperability, adoption, competition, and risk. Section V examines broader domestic economic and financial implications. Section VI presents comparative considerations for system design, and Section VII concludes.

II. Analytical Framework and Global Context

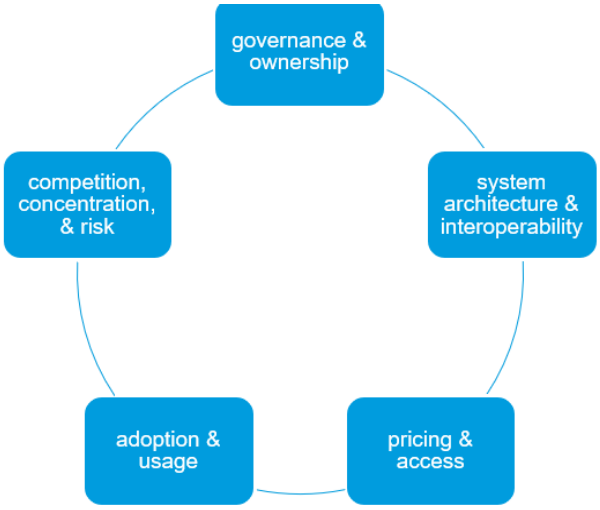
FPS should be understood not simply as payment technologies, but as institutionally embedded infrastructure whose performance depends on how governance, technical design, pricing, and access interact. A comparative framework is therefore needed to assess FPS not only by speed, but also by the institutional conditions that shape scale, usability, inclusion, and resilience.

A. Analytical Dimensions for Comparing Fast Payment System Models

The framework distinguishes between three institutional design dimensions that policymakers can influence directly and three outcome dimensions through which ecosystem performance is evaluated (Figure 1). The framework builds on a growing policy and empirical literature that treats FPS as institutional and ecosystem reforms rather than narrowly technical payment upgrades. The World Bank's fast-payment implementation work identifies policy objectives, governance arrangements, access rules, interoperability, risk management, and use-case development as core elements in FPS design (World Bank, 2021, 2022). BIS research similarly emphasizes that adoption and impact depend on design characteristics such as public-sector involvement, participation by banks and nonbanks, pricing incentives, use cases, and interoperability, while CPMI work on FPS interlinking highlights governance, oversight, legal setup, stakeholder engagement, scalability, and risk allocation as key analytical dimensions (BIS, 2023; CPMI, 2024). These precedents motivate the paper's comparative framework, while this paper adapts them to distinguish more explicitly among three institutional models: a unified public rail, a hybrid multi-rail structure, and a layered infrastructure/platform-led ecosystem.

1. **Governance and ownership**. This dimension examines whether the payment system is led by the central bank, operated through a public-private arrangement, or driven primarily by private platforms. Governance arrangements shape incentives for participation, standards setting, pricing structures, and access conditions. They also influence the extent to which public policy objectives—such as financial inclusion, competition, efficiency, and resilience—are embedded in system design and operation.
2. **System architecture and interoperability.** This dimension assesses whether the FPE is built around a single payment infrastructure, multiple coexisting infrastructures, or centralized back-end infrastructure with competing front-end schemes and applications. The number of infrastructures does not, by itself, determine interoperability. Interoperability depends on whether schemes and PSPs operate under compatible rules, technical standards, directories, and participation arrangements that allow payments to flow across providers and schemes.
3. **Pricing and access.** Pricing and access arrangements are key determinants of adoption by consumers and merchants. FPS generally achieve scale more rapidly when transaction costs for end users are low or negligible, merchant onboarding is

Figure 1. Analytical Framework for Comparing Fast Payment Systems



streamlined, and participation is open to a broad range of banks and nonbank payment service providers. Features such as proxy identifiers, QR-code functionality, and mobile-first interfaces can further reduce transaction frictions and enhance usability.

4. **Adoption, usage, and usability.** This outcome dimension evaluates the extent to which FPEs become embedded in everyday payment behavior across households, businesses, and government entities, and whether users can access and use them easily across routine use cases. Widespread institutional participation does not necessarily translate into intensive usage; successful ecosystems must become convenient and trusted payment options for routine transactions.
5. **Competition, concentration, and innovation.** This outcome dimension considers how different FPE models affect market structure, contestability, provider entry, and innovation, including whether network effects broaden competition at the service layer or reinforce concentration and systemic dependence.
6. **Trust, resilience, and broader policy outcomes.** This outcome dimension considers fraud, consumer protection, operational resilience, and confidence in the ecosystem, as well as broader objectives such as financial inclusion, payment efficiency, and public financial management. These outcomes depend on institutional design but are also shaped by country conditions and the surrounding financial and digital ecosystem.

The framework therefore assesses how the three design dimensions shape scale, usability, trust, competition, inclusion, efficiency, and resilience, while recognizing that these outcomes are also influenced by external country characteristics.

B. Payment Systems, Payment Schemes, and Ecosystem Architecture

The paper uses *fast payment ecosystem* as the umbrella term for the full institutional arrangement. An FPE comprises two closely related components: (i) the payment system or settlement infrastructure through which transactions are cleared and settled; and (ii) the payment scheme, including the common rules, standards, participation requirements, functionalities, and user-facing applications through which fast payment services are provided. Under this terminology, Pix is the Brazilian instant-payment scheme, while the Instant Payment System (SPI) is its settlement infrastructure; FedNow and RTP are U.S. payment systems or rails operating alongside multiple schemes and applications; and China combines public-sector payment infrastructure with predominantly private-sector payment schemes and applications. Distinguishing these two components within each ecosystem helps ensure that the three cases are compared on a like-for-like basis and clarifies whether observed outcomes reflect the infrastructure layer, the scheme and application layer, or the interaction between them. Not every transaction initiated through an FPE application necessarily settles through an FPS: wallet-to-wallet transfers within the same platform may be recorded on the provider's internal ledger, while funding, withdrawal, or cross-provider transactions may use bank, card, ACH, or instant-payment infrastructures.

C. Global Patterns and Institutional Models

Recent global FPE development shows both convergence and divergence. There is clear convergence toward real-time availability, mobile-centric interfaces, simplified user authentication, and low-cost retail payments, but countries continue to differ markedly in the institutional pathways through which these outcomes are pursued. To make these differences analytically tractable, this paper distinguishes three broad institutional models: **the unified public model**, in which a public authority coordinates an interoperable scheme and supporting infrastructure for system-wide use; **the hybrid multi-infrastructure model**, in which public and private infrastructures coexist and retail payments are distributed across multiple schemes and interfaces; and **the layered infrastructure/platform-led model**, in which centralized public infrastructure supports retail scale

through dominant private platforms. These models are assessed against a common set of outcomes: adoption and usage, usability, interoperability, competition and concentration, trust and resilience, financial inclusion, payment efficiency, and public-sector applicability. This typology helps clarify that there is no single institutional template for FPE development, but that outcomes still depend systematically on how governance, interoperability, pricing, and access are aligned.

India's Unified Payments Interface (UPI) provides an important additional variation within this typology. UPI operates as an open, publicly coordinated protocol that allows competing bank and nonbank applications to connect to a common interoperable network. It therefore separates competition at the user-interface layer from interoperability at the scheme and infrastructure layers. Although India is not examined as a full case study, UPI reinforces the broader analytical point that a common public architecture need not imply a single consumer-facing application and can instead support competition among multiple service providers (NPCI, 2026).

D. Policy Relevance for Emerging and Developing Economies

This framework is particularly relevant for emerging and developing economies, where payment modernization is often pursued alongside broader objectives of financial inclusion, state capacity, and digital transformation. In these settings, FPS design choices may affect not only payment efficiency, but also market contestability, the structure of financial intermediation, the reach of government transfers, and the development of digital public infrastructure more broadly. In small states and low-capacity environments, however, fixed costs, weak institutional capacity, and limited digital ecosystems may constrain implementation. For such countries, the question is not merely whether to adopt FPS, but which institutional model is most consistent with local scale, regulatory capacity, and policy priorities.

E. Why Brazil, the United States, and China

Brazil, the United States, and China are selected purposively rather than as a representative sample. They illustrate three contrasting institutional configurations: a unified public scheme supported by public settlement infrastructure; a hybrid model with multiple public and private infrastructures; and a platform-led ecosystem operating over centralized public infrastructure. The cases also differ in financial-sector structure, legacy payment use, regulatory capacity, and digital-platform development, allowing the analysis to identify recurring mechanisms and tradeoffs across markedly different settings. The three cases are not sufficient to identify the causal effect of individual design features. The paper therefore treats the comparison as structured institutional analysis and draws on broader empirical literature to assess specific mechanisms. For example, Pix adoption likely reflected the joint influence of mandatory participation, low user costs, simple aliases, merchant usability, coordinated supply-side integration, and network effects; more gradual U.S. adoption occurred against a different starting point, including mature card and ACH systems, established rewards programs, decentralized participation, and heterogeneous provider incentives. Differences in the speed and extent of FPS adoption are therefore not interpreted as evidence of the relative performance or efficiency of the underlying payment systems.

III. Country Case Studies

This section examines Brazil, the United States, and China as three distinct institutional models of fast payment development, focusing on governance, architecture, adoption, and tradeoffs.

A. Brazil: Pix as a Unified Public Rail Model

Brazil provides the clearest example of a unified public model, in which the central bank established the Pix scheme and operates the supporting SPI settlement infrastructure to advance interoperability, competition, and broad-based adoption.

1. Governance and Institutional Design

Brazil's Pix illustrates a centralized public-sector model in which the central bank combined regulatory authority, system design, and operational coordination to build a unified national fast payment infrastructure. Launched in November 2020, Pix formed part of a broader effort to modernize retail payments, reduce reliance on cash and legacy transfer instruments, strengthen competition in a concentrated banking system, and expand access to low-cost digital payments (Banco Central do Brasil, 2020; IMF, 2023).

A defining feature of Pix is that interoperability and broad reach were embedded from the outset. Under BCB Resolution No. 1 of August 12, 2020, participation became mandatory for financial and payment institutions with more than 500,000 active customer accounts, while smaller institutions could join voluntarily (Banco Central do Brasil, 2020). This accelerated network formation and reduced the risk that instant payments would emerge as fragmented proprietary schemes. At the same time, the framework allowed banks, digital banks, fintechs, and payment institutions to connect to the same infrastructure under common technical and operational rules, supporting competition at the service layer rather than through separate rails (World Bank, 2022; IMF, 2023).

2. System Architecture and Access Model

Pix is the instant-payment scheme that establishes the common rules, standards, participation requirements, and functionalities for real-time account-to-account payments. Its core settlement infrastructure is the Instant Payment System (SPI), which operates continuously and settles in central bank money, while the Directory of Transactional Account Identifiers (DICT) enables the use of simplified aliases—known as Pix keys—such as mobile phone numbers, email addresses, taxpayer identification numbers, or random keys instead of full account details (Banco Central do Brasil, 2020; World Bank, 2022). This architecture materially reduced user friction and helped make instant account-based payments viable for everyday retail use.

The access model was equally important to Pix's rapid scaling. Retail transfers for individuals were priced at zero in most use cases, while merchant acceptance costs were typically below those of card-based payments. The system also supported QR-code payments and alias-based transfers across a wide range of transactions, including person-to-person, person-to-business, business-to-person, and government-related payments (World Bank, 2022; IMF, 2023). By reducing user-side and merchant-side frictions, Pix achieved a degree of functional simplicity that many faster payment systems have struggled to attain.

3. Adoption, Usage, and Market Development

Pix adoption was exceptionally rapid by domestic and international standards. Within two and a half years of launch, the system had reached more than 140 million individuals and 13 million firms, according to BCB statistics cited by the IMF (IMF, 2023). By end-2024, Pix had become Brazil's leading retail payment instrument by transaction count: Brazilian banking industry data drawing on BCB statistics report 68.7 billion Pix

transactions in 2024, up 52 percent from the previous year, with transaction value of roughly US$5 trillion (FEBRABAN, 2025; Banco Central do Brasil, 2025). User penetration was also high, with Pix used by 76.4 percent of Brazil's population, compared with 69.1 percent for debit cards and 68.9 percent for cash (Banco Central do Brasil, 2025). More recent BCB Pix Statistics show continued expansion in users, registered accounts, transaction volumes, and values through 2025 and 2026, suggesting that Pix has remained the main growth segment of Brazil's retail payment system (Banco Central do Brasil, 2025–26).

These numbers make clear that Pix moved beyond early-adopter use into mass retail behavior. In 2024 alone, the system processed more transactions than the combined total of credit cards, debit cards, boletos, TED transfers, prepaid cards, and checks reported by the Brazilian banking federation. The scale of daily use was also striking: on December 20, 2024, Pix reportedly handled 252.1 million transactions in a single day, illustrating the extent to which it had become embedded in routine household, merchant, and business payments (FEBRABAN, 2025; Banco Central do Brasil, 2025).

Market development under Pix has also been notable in competitive terms. Because all participating institutions connect to the same rail, smaller banks, digital entrants, and fintech firms can offer instant payment services without having to replicate proprietary infrastructure or depend on incumbent card networks. This appears to have lowered barriers to entry and increased competitive pressure in Brazil's broader payments ecosystem, even though the banking system remains structurally concentrated (IMF, 2023; World Bank, 2022).

4. Main Tradeoffs and Constraints

Some external stakeholders have raised concerns that the BCB's combined regulatory and operational roles could create potential conflicts of interest, and that participation and application-visibility requirements could favor Pix over competing payment services (Campos, 2026). Brazilian authorities maintain that participation is open and access is non-discriminatory (Aguiar, Sant'Ana, and Taiar, 2026). These concerns are reported as external views rather than as findings of this paper, which does not adjudicate the competing claims.

Despite its success, Pix has also highlighted the operational and policy challenges associated with scaling instant payments rapidly. The most immediate concern has been fraud and cyber-related risk. The speed and irrevocability of transactions reduce the time available for intervention once a payment has been initiated, increasing exposure to fraud, social engineering, and account misuse. In response, the BCB has progressively strengthened the Pix rulebook, including transaction limits, device-based controls, reporting obligations, and procedures related to fraud signaling and security incidents (Banco Central do Brasil, 2023).

A second issue concerns the broader implications of instant payments for financial intermediation and liquidity management. Emerging research from the Banco Central do Brasil suggests that instant payments can increase banks' demand for liquid assets and reduce the scope for liquidity transformation by limiting the ability to delay and net payments (Gonzalez, Ma, and Zeng, 2025). While these effects do not detract from Pix's operational success, they suggest that the macro-financial implications of large-scale instant payments deserve more attention than is often given in policy discussions.

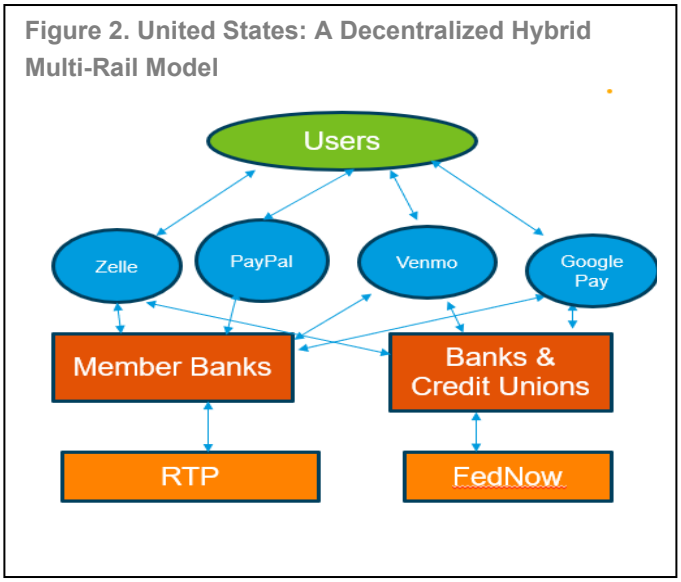
A third tradeoff relates to sustainability and market structure. Pix's low-cost model has delivered large benefits to users and merchants, but it can also compress fee-based revenue for payment service providers and intensify pressure on incumbent business models. Moreover, while Pix broadened competitive access to payments infrastructure, it has not fully eliminated concentration in Brazil's banking sector. The Brazilian case therefore shows that public digital infrastructure can increase contestability and lower transaction costs without necessarily resolving all structural issues in the surrounding financial system (IMF, 2023).

B. United States: A Decentralized Hybrid Multi-Rail Model

The U.S. represents a decentralized hybrid model in which public and private instant-payment rails coexist within a fragmented financial system alongside multiple consumer-facing payment platforms.

1. Governance and Institutional Design

The U.S. developed instant payments through a decentralized hybrid model in which public and private infrastructures coexist within a historically fragmented banking and payments system. Unlike countries that adopted a single national fast payment rail, the U.S. approach reflects a financial system with thousands of depository institutions, multiple clearing arrangements, and a longstanding private-sector role in payment modernization. In this setting, instant payments emerged through two parallel rails: the private-sector Real-Time Payments (RTP) network, launched by The Clearing House in 2017, and the FedNow Service, launched by the Federal Reserve Board in July 2023 (The Clearing House, 2025; Board of Governors of the Federal Reserve System, 2023).



This dual-rail structure reflects both market pragmatism and policy choice. The earlier launch of the RTP network allowed large banks and payment providers to develop real-time use cases before the Federal Reserve introduced its own service. The Federal Reserve Board has presented the FedNow Service as neutral public infrastructure intended to broaden access to instant payments across the banking system, including for smaller banks and credit unions, while supporting resilience and competition (Board of Governors of the Federal Reserve System, 2023). The result is a hybrid framework in which public and private rails operate in parallel rather than a single unified national architecture.

A further feature of the U.S. model is that consumer-facing digital payment activity is shaped not only by these core rails, but also by major front-end platforms and digital wallets. Applications such as Zelle, PayPal, Venmo, and Google Pay have familiarized users with rapid digital payments, but they are not themselves payment rails. Instead, they operate as overlays on existing bank and card infrastructures, with varying degrees of connection to instant-payment functionality. Applications such as Venmo may provide an immediate user experience even when the underlying transfer relies on an internal ledger, ACH, card networks, RTP, or another payment channel. This layering is central to the U.S. case: unlike Brazil, where the public scheme is also the dominant retail payment brand, the U.S. separates the infrastructure layer from much of the user-interface layer (Figure 2).

2. System Architecture, Rails, and Front-End Platforms

Both the RTP network and the FedNow Service provide 24/7/365 instant interbank payment capability based on credit-push architecture, immediate clearing, and near real-time settlement. The FedNow Service enables participating depository institutions to send and receive instant payments through Federal Reserve accounts and includes features such as liquidity management transfers, request-for-payment functionality, and tools to support fraud mitigation and payment inquiries (Board of Governors of the Federal Reserve System, 2023). The RTP network offers comparable real-time capabilities through privately operated infrastructure and has developed a broad set of business and consumer use cases over time (The Clearing House, 2025).

A defining characteristic of the U.S. system is therefore the coexistence of multiple core rails alongside multiple front-end platforms. At the infrastructure layer, the FedNow Service and the RTP network are the two principal instant interbank rails. At the user-facing layer, banks and nonbank providers offer various

applications for peer-to-peer transfers, digital wallet funding and defunding, mobile and in-app wallet payments, bill payment, payroll, and merchant-related transactions. This architecture differs from more unified models because the user experience depends heavily on which rail, institution, or application is being used. In practice, consumers may perceive payments as “instant” without knowing whether the transaction ultimately relies on an instant rail, a card-based mechanism, or another underlying channel.

3. Adoption, Usage, and Market Development

Adoption of instant interbank payments in the U.S. has been more gradual than in Brazil and China, but began from a different starting point, with sophisticated retail payment infrastructure and mature card and ACH alternatives already in place. Participation in the RTP network expanded steadily after its 2017 launch, and the FedNow Service has added a growing number of institutions since going live in 2023. By 2025–26, the FedNow Service had reached institutions in all 50 states, while the RTP network had more than 1,000 participating institutions and covered a large share of U.S. checking and savings accounts (The Clearing House, 2025; Federal Reserve Financial Services, 2026).

Usage patterns suggest that the strongest early traction for U.S. instant payments has emerged in selected business and account-based use cases rather than through uniform mass retail migration. Federal Reserve material points to growing interest in instant payroll, urgent disbursements, digital wallet funding and defunding, account-to-account transfers, and just-in-time business payments (Federal Reserve Financial Services, 2024, 2025). Public information on the RTP network likewise indicates rapid growth in business-to-business payments, gig-economy payouts, account-to-account transfers, and digital-wallet-related flows (The Clearing House, 2025). These patterns suggest that U.S. instant-payment adoption has been driven less by a single ubiquitous retail proposition than by a widening set of operational use cases across business and consumer segments.

At the same time, consumer familiarity with digital payments has been shaped more by front-end applications than by the direct visibility of the underlying rails. Survey evidence from the Federal Reserve and McKinsey & Company indicates that U.S. consumers increasingly use digital wallets and faster payment services, especially in mobile and in-app settings, while businesses value instant payments for cash-flow management and customer experience (Federal Reserve Financial Services, 2024; McKinsey & Company, 2024). Yet greater familiarity with digital payments does not by itself produce a single interoperable retail instant-payment ecosystem. Instead, the U.S. market remains distributed across card networks, ACH, bank-based applications, digital wallets, and the two core instant rails.

4. Main Tradeoffs and Constraints

The main tradeoff in the U.S. model is between institutional diversity and system-wide integration. The coexistence of the FedNow Service and the RTP network may strengthen resilience, competition, and provider choice, but it also creates coordination challenges. Interoperability remains limited not simply because two rails coexist, but because the relevant schemes, PSPs, directories, technical standards, and participation arrangements are not yet fully aligned across them. More broadly, legacy payment infrastructures, card dominance, and uneven participation incentives continue to slow migration toward instant account-to-account payments across the system.

A second constraint is that adoption incentives are not uniform across institutions. For large banks and service providers, the business case for instant payments may rest on commercial disbursements, treasury operations, account-to-account transfers, or competitive positioning. For smaller institutions, however, the costs of modernization, integration, and fraud management may weigh more heavily, especially when customer demand is still developing or can be met through other channels. This helps explain why participation has expanded gradually even after the launch of the FedNow Service.

A third tradeoff concerns fraud, consumer protection, and oversight in a layered ecosystem. As instant payments and digital wallet use expand, the speed and irrevocability of funds transfers can increase exposure to fraud, mistaken transfers, and disputes. These issues have drawn greater policy attention in the U.S., including through CFPB action to expand federal supervision over large digital payment apps and wallet providers, with particular emphasis on fraud, privacy, and consumer protection risks (Consumer Financial Protection Bureau, 2024). In this respect, the U.S. experience illustrates that a plural and innovative payment ecosystem may require more distributed and complex governance arrangements as instant-payment functionality scales.

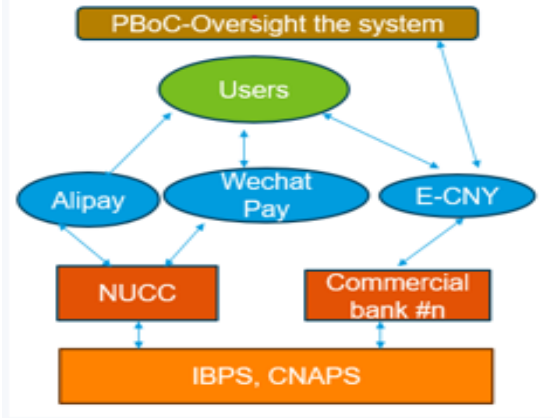
C. China: Centralized Back-End Infrastructure with a Platform-Led Retail Ecosystem

China represents a third model in which centralized public infrastructure supports a platform-led retail payment ecosystem dominated by large private wallets and increasingly complemented by e-CNY.

1. Governance and Institutional Design

China's fast-payment ecosystem combines centralized public infrastructure with platform-led retail innovation. Its core payment and clearing arrangements are overseen by the People's Bank of China (PBoC), including the Internet Banking Payment System (IBPS) within the broader China National Advanced Payment System (CNAPS). Unlike Brazil's Pix, however, China's retail payment transformation did not center on a single public-facing instant-payment brand. Instead, Alipay and WeChat Pay became the dominant user interface, while centralized public infrastructure remained largely in the background (Figure 3).

Figure 3. China: Centralized Back-End Infrastructure with a Platform-Led Retail Ecosystem



This configuration reflects a sequence in which market-led retail innovation expanded rapidly before regulatory centralization and oversight strengthened. During the 2010s, e-commerce, smartphones, a less-entrenched card-payment infrastructure, platform subsidies, and a relatively permissive early regulatory environment enabled Alipay and WeChat Pay to scale quickly (CGAP, 2019; Brookings, 2020). As the platforms became systemically important, the PBoC tightened oversight of nonbank payment institutions, strengthened clearing arrangements, and emphasized consumer protection, competition, and financial stability (IMF, 2024).

China's rapid adoption also reflected broader structural and institutional conditions. Unlike in the United States and parts of Europe, card-payment infrastructure was not deeply entrenched, reducing switching costs for consumers and merchants as smartphones and e-commerce expanded. Alipay initially addressed an important trust constraint in online commerce through an escrow arrangement that held funds until buyers confirmed receipt, helping establish confidence in digital transactions. Platform subsidies—including low or zero merchant fees, consumer promotions, and cash-back incentives—further encouraged QR-code acceptance and behavioral change. Rapid growth was also supported by a relatively permissive early regulatory environment, during which large technology firms expanded

payment networks and adjacent financial services before oversight was subsequently strengthened. The 2025 China FSAP underscores the continuing importance of ensuring that regulation and supervision remain commensurate with the scale, complexity, and interconnectedness of financial institutions and technology-led financial activities (IMF, 2025).

Institutionally, China is neither a purely public nor a purely private-sector model. Centralized public infrastructure provides the clearing and settlement backbone, while private platforms dominate consumer interaction, merchant acceptance, and ecosystem innovation. As Figure 3 illustrates, the NetsUnion Clearing Corporation (NUCC) occupies a pivotal middle layer between nonbank payment platforms, commercial banks, and the PBoC-supervised infrastructure. By routing and clearing transactions involving nonbank payment institutions through a centralized platform, NUCC reduced the platforms' reliance on bilateral connections with banks and strengthened regulatory visibility over retail payment flows. This architecture preserved platform-led innovation at the front end while bringing clearing more firmly within a standardized and supervised framework. The resulting model achieved scale but also created dependence on a small number of firms, prompting stronger oversight as the ecosystem matured (IMF, 2024; PBoC, 2021).

2. System Architecture and Platform Ecosystem

China's payment architecture is organized as a layered system in which centralized interbank infrastructure supports highly scalable front-end wallet platforms. At the infrastructure level, CNAPS provides the national payment backbone, while IBPS supports internet- and mobile-based retail payment clearing across participating institutions. Publicly available descriptions from the World Bank classify IBPS as a PBoC-operated system designed to support retail-oriented transfers using channels such as internet banking, mobile applications, and QR-code-linked use cases.

At the retail layer, Alipay and WeChat Pay provide the dominant user-facing wallet ecosystems linked to bank accounts and embedded in broader digital platforms. Some transactions may be completed through internal platform-ledger entries, while others involve external clearing or settlement arrangements. Because a reliable public breakdown between these transaction types is not available, the paper treats these platforms as part of the broader FPE rather than as FPS infrastructure themselves. Transactions are typically initiated through QR-code scanning, in-app payments, or embedded platform services. This architecture differs from the U.S. model because the front-end applications form the primary retail interface for everyday payments across commerce, transport, food delivery, and social interactions. In that sense, China's model places far more of the payment experience inside platform ecosystems than either Brazil or the United States (CGAP, 2019; Brookings, 2020).

A key enabling feature of this system has been QR-code-based acceptance. Relative to card-based acceptance infrastructure, QR codes substantially reduced merchant onboarding costs and enabled even very small merchants to accept digital payments with minimal hardware requirements. The World Bank notes that China's experience played an important role in demonstrating the scalability of QR-code payments in retail environments, while BIS research emphasizes that QR-code-based mobile payments generated a digital transaction footprint that could support broader access to financial services, especially for smaller firms (World Bank, 2021; BIS, 2022).

The institutional treatment of e-CNY changed materially on January 1, 2026, with the introduction of the upgraded e-CNY management framework. Under this framework, balances held in commercial-bank e-CNY wallets are treated as deposit liabilities of the operating banks, earn interest, receive deposit-insurance protection, and are incorporated into banks' asset-liability management and reserve-requirement calculations. Nonbank payment institutions must maintain 100 percent reserve backing for the e-CNY balances they

manage. The change moves e-CNY from its earlier cash-like design toward a deposit-based digital-money model under a tiered operating framework (People's Bank of China, 2026).

3. Adoption, Usage, and Market Development

China's digital payment adoption has been among the fastest and most extensive globally. Mobile payments became deeply embedded in everyday economic life across households, merchants, and service providers, with Alipay and WeChat Pay achieving near-ubiquitous visibility in many urban retail settings and broad penetration beyond them (CGAP, 2019; Brookings, 2020). By mid-2024, about 954 million individuals were reported to be using mobile payment services, and by end-2024 online payment users had exceeded 1.0 billion, equivalent to 92.8 percent of China's internet users (CNNIC, 2025; Daxue Consulting, 2024). These figures underscore that China's platform-led model reached population-scale adoption rather than remaining a niche digital-payment channel.

Merchant adoption was a particularly important driver of scale. Low-cost QR-code acceptance allowed many small merchants, informal vendors, and service providers to participate in digital payments without relying on traditional point-of-sale infrastructure. Complementing this broad user base, market estimates indicate that Alipay and WeChat Pay together accounted for more than 90 percent of China's mobile payment market, with Q3 2023 transaction values reported at RMB 118.19 trillion for Alipay and RMB 67.81 trillion for WeChat Pay (Daxue Consulting, 2024). This concentration helped deliver strong network effects and everyday usability, while also reinforcing the market-structure concerns discussed below. Alok et al. (2024) find that cross-platform digital payments can help generate transaction information that supports access to credit, suggesting that payment data may extend the ecosystem's relevance beyond payments alone.

At the same time, market development became highly concentrated. Although China's digital payments ecosystem is technologically sophisticated and broadly used, retail activity became increasingly dominated by the two largest platforms. This concentration supported strong scale and user convenience, but it also reduced contestability at the retail interface and increased the systemic importance of a small number of firms (IMF, 2024; Brookings, 2020).

4. Main Tradeoffs and Constraints

China's main tradeoff is between platform-driven scale and concentration risk. Centralized infrastructure and powerful private ecosystems enabled rapid adoption, low-friction transactions, and deep integration into everyday commerce, but also created dependence on a few dominant platforms, raising concerns about market power, data concentration, and systemic importance (IMF, 2024; Brookings, 2020).

A second issue is governance. As payments became embedded in broader fintech ecosystems, oversight had to extend beyond payment rails to platform supervision, nonbank payment regulation, clearing centralization, antitrust, AML obligations, and data governance. This suggests that platform-led payment systems require broader and more adaptive governance than conventional bank-centered models.

A third constraint concerns interoperability and inclusion at the margin. Although China achieved high consumer usability, competition between platform ecosystems created some segmentation, while digital payment reliance raised access concerns for some elderly and rural users. These issues prompted greater attention to standardization, interoperability, and inclusive payment options, alongside e-CNY as a publicly sponsored digital-money framework (IMF, 2024).

IV. Cross-Country Comparative Analysis

This section compares the three models through the main mechanisms identified in the analytical framework: governance, interoperability, pricing and access, competition, and risk. The purpose is not only to contrast country experiences, but to show how different institutional configurations shape network effects, adoption, and oversight tradeoffs.

A. Comparative Overview of the Three Models

Table 1 compares three institutional models across equivalent layers of their payment ecosystems: scheme rules, underlying payment systems or settlement infrastructure, and user-facing platforms. Brazil combines the unified public Pix scheme with the public SPI settlement infrastructure; the United States relies on the FedNow and RTP payment systems alongside multiple schemes and front-end platforms; and China combines public-sector payment infrastructure with predominantly private-sector payment schemes and platforms. Because publicly available metrics are not harmonized across the three ecosystems, the table uses selected indicators of usage and scale rather than a single narrowly defined measure. Some jurisdictions report transaction volumes or values for the underlying infrastructure, while others report users, participating institutions, merchant coverage, or usage indicators for particular schemes and applications. These indicators are therefore intended to provide a broad sense of reach and payment activity rather than a standardized quantitative ranking. Evaluated against a common set of criteria, the models reveal distinct tradeoffs. Brazil's unified public model scores strongly on interoperability, merchant usability, and speed of adoption when governance, pricing, and access are aligned, but it also places greater operational and fraud-management responsibility on the public sector. The U.S. hybrid multi-rail model offers greater provider diversity and may support resilience and innovation, but it tends to generate slower and less uniform retail diffusion because users and merchants interact with multiple rails and front-end platforms. China's layered infrastructure/platform-led model can achieve rapid retail scale through ecosystem coordination and low-friction merchant acceptance, but it also raises stronger concerns about concentration, data governance, and oversight complexity.

The typology also helps sharpen the paper's policy relevance for emerging and developing economies. A unified public rail model may be especially attractive where central-bank capacity is strong and policymakers want to build interoperability and broad access from the outset. A hybrid multi-rail model may be more feasible in fragmented financial systems with strong private-sector payment innovation, but it requires stronger coordination mechanisms than is often assumed. A layered infrastructure/platform-led model may achieve rapid scale where large digital platforms are already deeply embedded, but it also demands robust ex post regulatory capacity to manage concentration and ecosystem dependence. The subsections below examine these tradeoffs in more detail across governance, interoperability, adoption, competition, and oversight.

B. Governance and Institutional Design

The three cases differ fundamentally in governance, and those differences help explain variation in interoperability, adoption, and market structure. Brazil adopted the most centralized model: the central bank designed, regulated, and coordinated Pix as a unified national infrastructure, embedding interoperability, access, and pricing objectives directly into the system while allowing competition to develop around a common rail. This mattered not only because the system was public, but because governance was used to align participation, standards, and incentives from the outset. The U.S. adopted a more decentralized arrangement in which FedNow and RTP coexist within a fragmented financial system, supporting resilience and provider choice but weakening centralized coordination and slowing system-wide network effects. China represents a

layered model in which public authorities control the core clearing and settlement backbone while private platforms dominate much of the retail interface, merchant integration, and innovation dynamic, allowing rapid scale through ecosystem coordination rather than through a single unified rail.

C. System Architecture and Interoperability

Architecture and interoperability help explain why these ecosystems scaled differently. In Brazil, Pix is interoperable by design because participating PSPs operate under a single scheme with common rules, standards, identifiers, and open participation, supported by the SPI settlement infrastructure. In the United States, FedNow and RTP provide multiple payment infrastructures, but the existence of multiple rails does not itself preclude interoperability. Rather, interoperability remains more limited because the relevant schemes, PSPs, directories, and participation arrangements are not yet fully aligned. In China, centralized infrastructure supports broad retail usability through dominant wallet ecosystems, although interoperability across competing schemes and platforms has historically been less complete.

These differences illustrate that interoperability should be assessed at the scheme and provider levels, rather than inferred solely from the number of payment rails. A single open scheme may be interoperable by design, as in Pix, while multiple infrastructures can also support interoperability if they operate under compatible rules, technical standards, directories, and participation arrangements. In China, interoperability has been strong in functional retail terms, but historically less complete across competing platforms, prompting later efforts to centralize clearing and improve standardization. The introduction of e-CNY adds a further public layer, but not a direct substitute for existing infrastructures or wallets.

D. Pricing, Access, and Adoption Dynamics

Pricing, access conditions, and user friction help explain why adoption was much faster in Brazil and China than in the United States. Technical capability alone did not generate scale. What mattered was whether instant payments were inexpensive for users, easy for merchants to accept, broadly available across institutions, and embedded in low-friction interfaces. In Brazil, rapid diffusion reflected the combination of mandatory participation by large institutions, zero-cost retail transfers, common proxy identifiers, and broad merchant usability. In China, fast scale reflected low-cost QR-based acceptance, platform embedding, and mobile-first behavioral integration. In the United States, by contrast, adoption has been slower and more segmented because instant-payment capability is distributed across multiple rails, coexists with well-established legacy alternatives, and lacks a similarly unified merchant and consumer value proposition.

Brazil achieved the strongest alignment between low pricing, broad access, and adoption. Pix combined mandatory participation by large institutions, zero-cost retail transfers in most use cases, simplified proxy identifiers, low merchant acceptance costs, and broad interoperability across institutions. Because the system was immediately usable for peer-to-peer, merchant, and government-related payments, network effects reinforced one another quickly and instant payments became a general-purpose retail instrument.

China appears to have reached comparable retail scale through a different mechanism. QR-code payments sharply lowered merchant onboarding costs, while integration with widely used digital platforms made payment functionality part of existing commercial and social routines. Broad access thus came less through a single public-facing rail than through the ecosystem reach of platform-based payments and the mobile-first habits that sustained repeated use. e-CNY adds a public digital option, but so far complements rather than displaces the dominant private channels.

The United States shows that instant-payment capability does not automatically produce mass retail adoption when pricing, access, and user experience remain fragmented. FedNow and RTP have expanded real-time transfer capacity, but they coexist with entrenched legacy instruments, card-based consumer habits,

and front-end platforms such as Zelle, PayPal, Venmo, and Google Pay that do not amount to a single interoperable retail proposition. As a result, the strongest traction has come in selected business, payroll, account-to-account, and disbursement use cases rather than through unified retail migration.

E. Competition, Concentration, and Innovation

The three models also organize competition differently. In Brazil, competition occurs mainly around a common public rail, with firms differentiating through customer experience and adjacent services rather than proprietary infrastructure. In the U.S., competition is spread across infrastructures, institutions, and front-end platforms, which supports experimentation but also segmentation. In China, rapid innovation initially came through large private platforms, but scale later raised concentration concerns and prompted stronger public intervention. The broader lesson is that instant-payment design shapes not only adoption, but also where competition occurs and whether innovation strengthens cohesion or fragmentation.

F. Risk, Oversight, and Institutional Tradeoffs

As fast payment systems scale, the policy challenge extends beyond efficiency to fraud, resilience, consumer protection, and systemic dependence. These risks differ across models. Brazil's unified rail allows coordinated system-wide responses but concentrates operational and fraud-management responsibilities. The U.S. distributes risk across multiple rails, platforms, and regulators, which can support redundancy but complicates oversight. China's main challenge stems from concentration within large platform ecosystems, requiring a broader regulatory perimeter that extends beyond payments narrowly defined. Risk management is therefore integral to system design, not a separate layer added after scale is reached.

Brazil's unified public model concentrates operational scale and fraud-management responsibilities within a common national infrastructure. This supports clear governance and rapid system-wide response, but high Pix volumes have also increased exposure to fraud associated with irreversible instant transfers, prompting tighter monitoring, transaction limits, and stronger fraud-prevention measures. Brazil thus shows how centralized design can strengthen coordination while placing greater pressure on the public operator to maintain security, resilience, and trust.

The U.S. faces a different oversight tradeoff because risk is distributed across multiple rails, platforms, and regulatory authorities. This may support redundancy, but it also complicates coordination and can create uncertainty for users when fraud, mistaken transfers, or disputes arise. The growing policy focus on large digital payment apps and wallet providers reflects recognition that instant and quasi-instant payments raise not only operational issues, but also broader consumer-protection and data-governance concerns.

China's main oversight challenge stems from the concentration of payment activity within a small number of platform ecosystems. This structure appears to have delivered exceptional scale and usability, but also increased concerns about market power, data concentration, and systemic dependence. Regulatory attention therefore expanded beyond payments narrowly defined to include centralized clearing, nonbank supervision, antitrust oversight, and broader fintech regulation. In that context, e-CNY also helps preserve a public-policy role in the architecture of retail digital payments.

G. Comparative Synthesis

The comparative evidence suggests that FPS outcomes depend less on public versus private provision than on whether governance, interoperability, pricing, and access are aligned to support scale, usability, and trust. Institutional design shapes where network effects accumulate, how competition is organized, and which oversight burdens emerge as systems scale.

Shared infrastructure tends to concentrate network effects and support contestability at the service layer; plural rail structures tend to diffuse innovation across providers but weaken unified migration; and platform-led ecosystems can scale rapidly through merchant and user integration, while creating stronger concentration and governance risks. These tradeoffs are central to how FPS models should be evaluated.

There is no single optimal institutional model for FPS development, but some design principles are more consistently associated with successful outcomes: strong coordination capacity, broad interoperability, low-friction access, and governance frameworks that evolve with the system's market structure and risk profile. The next section considers how these institutional differences translate into broader economic and financial implications.

Table 1. Major Instant Payment Platforms: Comparative Analysis

Jurisdiction / System	Launch Year	Ownership / Governance	Selected indicators of usage and scale	Annual / Cumulative Transactions	Main Identifiers Used
Brazil – Pix	2020	Operated by the Central Bank of Brazil as public instant payment infrastructure	Over 180 million users by 2025; now Brazil's dominant retail payment method	Approximately 79.8 billion transactions and BRL 35 trillion in 2025	Phone number, email, CPF/national ID, random key, QR code
United States – Zelle	2017	Owned by Early Warning Services, a consortium of major U.S. banks	151 million enrolled accounts in 2024	US$1 trillion sent across 3.6 billion transactions in 2024	Mobile number, email address linked to bank account
U.S. – PayPal	1998	Operated by PayPal	More than 430 million active accounts globally in 2024	Around US$1.7 trillion total payment volume in 2024	Email address, phone number, username/account ID, QR code
U.S. – Venmo	2009	Owned by PayPal	Roughly 90 million active users in the U.S.	Estimated US$286–290 billion transaction volume in 2024	Phone number, username, QR code
U.S. – Google Pay	2011	Operated by Google (Alphabet Inc.)	Approximately 35–38.8 million active users in the U.S. by 2025	Over 2.1 billion payments processed globally in 2025; U.S. market share at 61.9%	Email address, phone number, QR code, NFC tap-to-pay
China – WeChat Pay	2013	Operated by Tencent through Tenpay under PBoC oversight	Estimated 900+ million active users integrated into the WeChat super-app ecosystem	Together with Alipay, estimated to process more than RMB 300 trillion annually	QR code, phone number, WeChat ID, merchant QR/token
China – Alipay	2004	Operated by Ant Group under PBoC regulation	Estimated 800+ million active users in China	Together with WeChat Pay, annual transaction flows estimated above RMB 300 trillion	QR code, phone number, Alipay ID, biometric/tap identifiers
China – e-CNY	Pilot launched in 2020	Issued by the PBoC under a two-tier CBDC model involving commercial banks	Pilots expanded across major cities and provinces with more than 100 million wallets reported	Approximately RMB 16.7 trillion cumulative transaction value and 3.48 billion cumulative transactions by late 2025	Digital wallet ID, phone number, QR code, hardware wallet/token

Sources:
Pix: Usage and transaction statistics are based on disclosures and reporting from the Central Bank of Brazil and major media summaries.
Zelle: User and transaction figures are from official disclosures by Zelle and reporting by CNBC.
PayPal and Venmo: Statistics are derived primarily from PayPal Investor Relations annual disclosures and financial reporting.
Google Pay: User statistics and transaction figures are based on Google/Alphabet official disclosures, industry analyses, and financial reporting from sources such as Statista and Business of Apps.
WeChat Pay and Alipay: Official transaction data are limited. Estimates are compiled from industry analyses, Reuters reporting, Ant Group disclosures, Tencent filings, and market intelligence reports.
e-CNY: Transaction and wallet figures are based on publications and speeches by the People's Bank of China and related policy reporting.
Notes:
Usage and scale indicators reflect the most informative publicly available measures for each ecosystem. Because reporting practices, institutional coverage, "Cross-border capabilities" refers to operational payment functionality and interoperability arrangements rather than full capital account convertibility or unrestricted international use.

V. Economic and Financial Implications

This section examines how the three models translate into broader economic outcomes, focusing on inclusion, efficiency, competition, public financial management, resilience, and digital transformation.

A. Financial Inclusion and Payment Access

Fast payment systems can strengthen inclusion by lowering transaction costs, increasing the practical value of transaction accounts, and reducing merchant acceptance barriers. The three cases suggest that these gains are strongest where instant payments are inexpensive, easy to use, and embedded in everyday transactions. In that sense, FPS support inclusion not only by expanding access to accounts, but also by increasing the frequency and usefulness of digital financial participation.

Brazil provides an instructive case of broad-based inclusion through a unified public payment infrastructure. Pix combined universal interoperability, zero-cost retail transfers in most use cases, simplified proxy identifiers, and broad participation, lowering barriers for households, small merchants, and informal workers. Evidence on Brazil and other Latin American systems indicates that low adoption costs, network effects, and coordinated supply-side integration can broaden practical use across users and firms rather than merely shift activity across electronic channels (Argente et al., 2025).

China appears to have achieved strong inclusion effects through platform-based usability and low-cost merchant access. QR-code acceptance and integration into daily digital activities allowed large numbers of consumers and small merchants to participate in electronic payments with limited upfront cost. At the same time, inclusion through private platforms coexisted with concentration and access challenges for some user groups, including parts of the rural and elderly population.

The U.S. shows more gradual and segmented inclusion effects. FedNow and RTP expand the infrastructure for real-time account-to-account payments, but their inclusion impact is moderated by high banking penetration, continued reliance on cards and legacy channels, and a fragmented retail payments ecosystem. Early benefits have therefore emerged mainly in specific use cases such as payroll, disbursements, and account-to-account transfers, improving the quality and timeliness of payment access more than formal inclusion itself.

B. Payment Efficiency, Transaction Costs, and Liquidity

A central economic effect of fast payment systems is to reduce settlement frictions, lower transaction costs, and improve cash-flow timing. In this section, efficiency refers to transaction costs, payment speed and predictability, and cash-flow and liquidity management; the assessment is qualitative and does not directly measure consumer welfare or economy-wide resource allocation. In all three cases, FPS development expanded the capacity for faster and more predictable payments, but the size of the gains depended on how widely instant payments were adopted, how fully they displaced legacy instruments, and whether system design supported low-cost everyday use. Speed matters most when it changes transaction costs, liquidity management, and operational efficiency across households, firms, and financial institutions. The comparison does not isolate the effect of rail structure or establish that unified infrastructure is inherently more efficient than competing rails.

Brazil illustrates broad-based efficiency gains associated with low-cost payments and widespread adoption, rather than evidence of the superiority o a unified rail. By offering free or low-cost instant payments with broad reach, Pix reduced reliance on higher-cost card and legacy transfer instruments, lowered merchant acceptance costs, and simplified payment initiation. Efficiency gains therefore extended beyond faster

settlement to lower transaction costs, reduced payment uncertainty, and smoother cash-flow management for households, firms, and merchants.

China appears to have achieved efficiency gains through centralized interbank infrastructure combined with deeply integrated platform ecosystems. QR-code wallet payments lowered merchant onboarding costs, reduced dependence on cash and traditional point-of-sale infrastructure, and embedded payments into e-commerce, transport, food delivery, and other digital services. In this setting, efficiency gains came not only from faster funds transfer, but also from integrating payments into broader digital workflows.

The U.S. shows a more differentiated efficiency profile. FedNow and RTP improve settlement speed, reduce payment delays, and strengthen liquidity management in time-sensitive use cases such as payroll, disbursements, account-to-account transfers, and business payments. These gains are meaningful, but broader retail efficiency gains depend on adoption, integration with legacy systems, and coordination across providers, rather than the number of rails alone. The available evidence does not establish a comparable ranking of efficiency gains across the three countries.

Liquidity effects also differ across the three systems. In Brazil, the success of Pix has drawn attention to the implications of continuous real-time settlement for bank liquidity management, including higher demand for liquid assets. Similar issues are more likely to matter in the United States in selected institutional use cases because adoption remains less universal. In China, liquidity effects are tied more closely to the high velocity and dense integration of retail payments within platform ecosystems, though their implications for banks are less visible because user interaction is heavily intermediated by nonbank platforms.

C. Market Structure and Financial Sector Competition

Fast payment systems can reshape market structure by changing how payment services are priced, how firms access core infrastructure, and where competition occurs. Their effects differ across models organized around a common interoperable rail, multiple competing rails, or dominant platform ecosystems. Payment infrastructure can therefore either broaden contestability or reinforce fragmentation and concentration, depending on how access and interoperability are designed.

Brazil illustrates how a common public rail can support competition by standardizing the infrastructure layer while allowing firms to differentiate at the service layer. Pix reduced dependence on proprietary card networks and legacy transfer instruments by giving banks, digital banks, and payment institutions access to the same interoperable backbone under common technical rules (Banco Central do Brasil, 2020; World Bank, 2022). This likely lowered entry barriers for smaller firms and fintechs, reduced the need to build proprietary payment rails, and increased competitive pressure on incumbents in payments and adjacent financial services (IMF, 2023). In this sense, Brazil's experience suggests that standardized public infrastructure can promote contestability when access is broad and interoperability is embedded from the outset.

The U.S. presents a different market-structure dynamic in which competition is distributed across multiple rails, institutions, and front-end applications. FedNow and RTP compete at the infrastructure layer, while banks, card networks, fintechs, and digital wallets compete on the user-facing layer. This supports innovation through diversity and experimentation, but because providers compete across fragmented channels rather than a common interoperable backbone, competition produces more service variety than system-wide convergence.

China shows how platform-led payment expansion can initially intensify innovation and competition, but later give rise to concentration at the retail interface. Alipay and WeChat Pay rapidly scaled merchant acceptance, consumer use, and adjacent services, but also concentrated a large share of transaction activity within a small number of firms. The authorities responded by strengthening centralized clearing, increasing oversight of nonbank payment institutions, and promoting e-CNY as a complementary public instrument.

D. Government Payments and Public Financial Management

Fast payment systems can improve public financial management by accelerating disbursements, reducing payment frictions, and extending the reach of digital public services. These benefits depend on how deeply instant payments are integrated into public-sector operations and administrative systems. The three cases suggest that FPS can support more timely and traceable government payments, but such gains are strongest where the system is interoperable, broadly accessible, and connected to existing public payment channels. Brazil provides a strong example of government payments benefiting from a unified instant-payment infrastructure. Pix has increasingly been used for public-sector collections, bill payments, and selected disbursement channels, reflecting its broad reach and low transaction costs. Its standardized and immediate functionality can improve the speed and traceability of transfers while lowering administrative frictions, showing how a public payment rail can strengthen state capacity when embedded in routine fiscal and quasi-fiscal transactions.

China also shows important links between digital payments and public-sector service delivery, though through a different institutional pathway. Government-related transactions increasingly occur within a broader ecosystem dominated by private platforms, while e-CNY creates additional possibilities for more direct and programmable public-sector transfers. For now, however, its role remains complementary because most routine digital payments still flow through Alipay and WeChat Pay.

The U.S. shows a more gradual and function-specific public-sector role for instant payments. FedNow and RTP create the technical conditions for faster government disbursements and improved timing of public payments, especially in payroll, emergency transfers, and account-based benefit delivery. But the broader public financial management impact remains more limited than in Brazil because instant-payment usage is not concentrated in a single rail with universal retail reach.

E. Financial Stability, Risk, and Operational Resilience

As fast payment systems become more embedded in everyday economic activity, their implications extend beyond efficiency to liquidity management, fraud risk, operational resilience, and systemic dependence. These risks do not arise uniformly across models. They depend on how instant payments are structured, who controls the infrastructure, and how concentrated the surrounding ecosystem becomes. Stability effects are therefore shaped as much by institutional design as by transaction speed.

Brazil illustrates both the strengths and the demands of a unified, high-volume public infrastructure. Pix has widened the speed and reach of transactions, but its rapid expansion has also increased exposure to fraud and cyber-related threats, requiring progressively stronger monitoring and controls. Gonzalez, Ma, and Zeng (2025) suggest that large-scale instant payments can affect bank liquidity management by increasing demand for liquid assets and reducing the scope for liquidity transformation.

The U.S. faces a distinct set of stability and resilience issues because risk is distributed across multiple rails, legacy systems, and user-facing platforms. This structure may provide redundancy, but it also complicates coordinated risk management across rails, applications, and regulators. In the U.S. context, financial stability concerns arise less from a single dominant rail than from the complexity of oversight in a plural ecosystem.

China’s resilience challenges are most closely linked to platform concentration and the broader embedding of payments within large digital ecosystems. The dominance of Alipay and WeChat Pay enabled extraordinary scale and convenience, but also concentrated payment activity and related data flows within a small number of firms. Regulatory responses therefore extended beyond narrow payments oversight to include centralized clearing, nonbank supervision, antitrust, data governance, and broader fintech regulation, while e-CNY helps preserve a direct public role in retail digital payments.

F. Digital Transformation and Wider Economic Effects

Beyond payment efficiency, fast payment systems can support wider digital transformation by changing how households, firms, and governments transact and organize economic activity. The three cases suggest that FPS can enable e-commerce, embedded finance, digital public services, and greater formalization of small-scale activity. The scale of these effects depends on how deeply instant payments are integrated into everyday commercial and administrative ecosystems.

Brazil shows how a publicly coordinated FPS can accelerate digitalization across retail commerce and financial services. Pix lowered frictions in person-to-person and merchant payments, reduced dependence on cash, and made digital transactions more accessible for households, small businesses, and informal workers. As adoption scaled, it supported wider digitization of retail activity and provided a common infrastructure on which financial institutions and fintechs could build complementary services.

China appears to show the strongest link between fast payments and wider digital transformation because payments became deeply embedded within platform ecosystems. Alipay and WeChat Pay helped reorganize commerce around mobile-first transactions spanning retail, transport, food delivery, social transfers, and embedded finance. In this setting, digital payments became part of a broader restructuring of economic activity rather than a narrow payments modernization exercise, while e-CNY adds a further public dimension to the evolving landscape.

The U.S. shows a more incremental but still meaningful relationship between instant payments and digital transformation. FedNow and RTP support modernization in selected business, payroll, treasury, and account-to-account use cases, but broader effects are less concentrated because the payments ecosystem remains distributed across multiple rails, card networks, and front-end applications. Instant-payment development therefore contributes more gradually to modernization than in Brazil or China.

More broadly, FPS function as enabling infrastructure whose economic impact depends on how they are embedded in the surrounding financial and digital ecosystem. Governance, interoperability, pricing, and access shape the channels through which benefits spread—whether through broader account use, lower transaction costs, improved payment timing, stronger merchant digitization, or more effective delivery of public payments.

G. Synthesis

Overall, the economic effects are strongest where the mechanisms supporting routine use are aligned. The key transmission channels are lower user and merchant friction, broader everyday usability, stronger merchant acceptance, and wider embedding of instant payments in commercial and administrative routines. Where these channels reinforce one another, FPEs can support inclusion, efficiency, government payment modernization, and wider digitalization. Where they remain partial or segmented, gains tend to diffuse more unevenly.

VI. Comparative Considerations for System Design

The comparison highlights several considerations that may be relevant when assessing alternative fast payment ecosystem designs. Their applicability will depend on country-specific institutional capacity, market structure, and policy objectives.

A. Governance, Interoperability, and Access

The cases suggest that governance, interoperability, and access are most effective when considered jointly at the design stage. FPE scale appears stronger when institutional responsibilities, participation rules, technical standards, and user-access arrangements reinforce one another from the outset. Where these elements remain fragmented, adoption tends to be slower and broader system benefits more uneven.

Brazil illustrates one design lesson especially clearly: interoperability is most effective when embedded at launch rather than layered on later. Pix aligned participation rules, common standards, simplified identifiers, and low-friction access from the outset, which helped accelerate network formation and reduce fragmentation (Banco Central do Brasil, 2020; IMF, 2023). The broader policy lesson is not to replicate Brazil's institutional model, but to treat interoperability as a foundational design choice rather than a later corrective.

The United States illustrates a different lesson: plurality requires stronger coordination than is often assumed. When multiple rails, legacy systems, and front-end platforms coexist, system-wide interoperability and uniform retail access are harder to achieve without common standards, broad participation incentives, and clear governance arrangements (Board of Governors of the Federal Reserve System, 2023; The Clearing House, 2025). The policy issue is therefore not whether hybrid models are viable, but how fragmentation can be limited when multiple public and private actors coexist.

China points to a third lesson: platform-led scale can substitute for formal interoperability, but only if governance keeps pace with concentration and ecosystem dependence. High retail usability can emerge through layered architecture and platform integration, but authorities may then need stronger ex post tools to preserve interoperability, maintain regulatory visibility, and limit systemic dependence on private platforms (People's Bank of China, 2021; IMF, 2024). The design lesson is to ensure that governance frameworks evolve with scale rather than assuming that rapid adoption alone will deliver balanced outcomes.

B. Adoption, User Friction, and Incentives

Broad adoption depends less on technical capability alone than on whether instant payments are simple, low-cost, and immediately useful in everyday transactions. Systems scale faster when retail users face minimal fees, merchants can be onboarded cheaply, and payment initiation is easy across institutions and devices.

Brazil illustrates how low friction and clear incentives can accelerate rapid system-wide adoption. Pix combined zero-cost retail transfers in most use cases, low merchant acceptance costs, proxy-based payment identifiers, and broad institutional participation, making the system immediately usable for households, merchants, firms, and public-sector entities (Banco Central do Brasil, 2020; IMF, 2023). These features reduced both user-side and merchant-side frictions and helped normalize instant payments as a general-purpose retail instrument. More broadly, Brazil suggests that adoption is strongest when pricing and access arrangements are aligned with a clear strategy to reach scale early in the system's lifecycle.

China reached similarly strong adoption through a different incentive structure centered on platform integration and low-cost merchant acceptance. QR-code payments reduced the fixed costs of merchant onboarding, while embedding payment functionality into widely used platforms made digital payments highly

convenient for consumers across everyday retail and service transactions (World Bank, 2021; BIS, 2022). In this setting, users did not need to adopt a new standalone payment behavior; instead, payment functionality was integrated into existing digital routines. The policy implication is that adoption incentives can also be created through user-interface design and merchant economics, not only through pricing policy at the core rail level.

The U.S. shows that adoption is likely to be slower and more segmented when pricing, access, and user experience remain distributed across multiple systems. FedNow and RTP expand instant-payment capability, but the coexistence of multiple rails, legacy payment instruments, and heterogeneous front-end platforms has limited the emergence of a single mass retail proposition (Board of Governors of the Federal Reserve System, 2023; Federal Reserve Financial Services, 2024; The Clearing House, 2025). This suggests that countries using hybrid or decentralized models may need stronger adoption incentives, including broad participation by major institutions, clearer merchant value propositions, and targeted public-sector use cases that help establish early scale.

C. Competition and Fragmentation

The cases suggest that competition at the service layer may be easier to sustain when fragmentation at the infrastructure layer is limited. Excessive fragmentation can weaken network effects and slow adoption, while platform-led scale can create concentration risks that reduce contestability and complicate oversight.

Brazil illustrates how shared infrastructure can support contestability when the service layer remains open. Standardizing the core rail and broadening access allowed firms to compete through customer experience and adjacent services rather than through separate proprietary payment rails (Banco Central do Brasil, 2020; IMF, 2023). The broader design lesson is that common infrastructure can strengthen competition when governance and access rules keep the service layer open.

India’s UPI similarly shows that common standards and infrastructure can coexist with competition among consumer-facing applications.

The United States highlights the opposite design risk: institutional diversity can coexist with persistent fragmentation. Where multiple rails and interfaces evolve without strong coordination, innovation may remain distributed across channels rather than generating common retail network effects (Board of Governors of the Federal Reserve System, 2023; The Clearing House, 2025). The policy lesson is to complement plurality with shared standards, interoperability protocols, and broad participation if the goal is system-wide diffusion.

China illustrates a third challenge, namely the risk that rapid platform-led innovation may evolve into concentration at the retail interface. Alipay and WeChat Pay generated powerful network effects and transformed digital payments at scale, but their success also concentrated transaction activity, data, and market power within a small number of firms (CGAP, 2019; People’s Bank of China, 2021). The subsequent strengthening of centralized clearing, nonbank payment oversight, and the development of e-CNY reflect efforts to preserve a greater public role and reduce systemic dependence on dominant private platforms (IMF, 2024). This suggests that authorities should monitor whether platform-based growth is reinforcing contestability or gradually narrowing it.

D. Risk Management, Oversight, and Consumer Protection

As fast payment systems scale, regulatory frameworks may need to evolve in parallel to address fraud, resilience, data governance, and consumer protection. These risks arise under all models, but in different forms: unified public rails concentrate operational responsibilities, fragmented systems complicate coordination, and platform-led ecosystems widen the regulatory perimeter and increase systemic dependence on a small number of firms. The comparison therefore treats risk management as a core design consideration rather than only a response after scale is reached.

Brazil illustrates the need for strong system-wide fraud controls and operational safeguards in a unified high-volume infrastructure. As Pix usage expanded rapidly, the Banco Central do Brasil introduced transaction limits, enhanced monitoring requirements, and strengthened fraud-prevention procedures to address fraud and social-engineering risks associated with irreversible real-time payments (Banco Central do Brasil, 2023). More broadly, the Brazilian experience suggests that centralized governance can support rapid system-wide response, but also places greater pressure on the operator to maintain trust, resilience, and service continuity as transaction intensity grows.

The U.S. highlights the oversight challenges of a layered ecosystem in which risks are distributed across multiple rails, institutions, and user-facing platforms. FedNow and RTP improve settlement speed, but they coexist with card networks, bank applications, and digital wallets that create a more complex consumer and supervisory environment (Board of Governors of the Federal Reserve System, 2023; The Clearing House, 2025). In this setting, consumer-protection issues such as fraud, mistaken payments, and dispute handling may be harder to manage consistently across channels. The CFPB's expanded oversight of large digital payment applications underscores the importance of clear supervisory coverage for nonbank providers as instant and quasi-instant payment functionality becomes more widespread (Consumer Financial Protection Bureau, 2024).

China illustrates how platform-led scale can require broader and more adaptive oversight. The concentration of payment activity within Alipay and WeChat Pay raised concerns not only about payments risk narrowly defined, but also about data concentration, market power, and the interaction between payments and broader fintech activities (People's Bank of China, 2021; IMF, 2024). Regulatory responses therefore extended beyond conventional payment oversight to include centralized clearing, nonbank supervision, antitrust measures, and broader fintech regulation. The development of e-CNY also points to a public-sector effort to preserve resilience and maintain a direct role in the retail digital payments landscape, even if its current role remains supplementary.

E. Institutional Capacity and Country Context

There is no single optimal FPE model; the relevance of different approaches depends on country capacity, market structure, and policy objectives. A central comparative consideration is which design best fits local regulatory reach, financial-sector structure, and the intended role of payments in broader economic development.

Countries with strong central-bank capacity, concentrated banking systems, and clear public-sector modernization objectives may be better positioned to adopt more centralized and interoperable models. Brazil's experience suggests that where authorities can coordinate participation, set common standards, and ensure broad access, a unified national rail can support rapid adoption, inclusion, and efficiency gains (Banco Central do Brasil, 2020; IMF, 2023). At the same time, such a model also requires sustained operational capability, fraud management, supervisory coordination, and the institutional credibility needed to maintain system-wide trust at scale.

Countries with more fragmented financial systems or stronger traditions of private-sector payment innovation may need to rely on hybrid arrangements, but these models require stronger coordination than is often assumed. The U.S. experience shows that plural structures can support innovation and provider choice, but may also generate fragmentation if governance, interoperability standards, and participation incentives are not aligned (Board of Governors of the Federal Reserve System, 2023; The Clearing House, 2025). In such settings, policy should focus not only on enabling infrastructure competition, but also on preserving system coherence through common standards, broad access rules, and effective supervisory coverage across institutions and platforms.

Platform-led ecosystems can scale rapidly where digital platforms are deeply embedded, but they are not a low-capacity shortcut. China's experience shows that low merchant onboarding costs and integration into

existing digital routines can accelerate adoption (CGAP, 2019; People's Bank of China, 2021). Yet such models require strong ex post regulatory capacity to manage concentration, data governance, systemic dependence, and an expanding oversight perimeter; where supervisory capacity is weak, risks may outpace regulation.

For small states and low-capacity economies, a phased or shared approach may be more appropriate than a fully standalone national system. Limited scale, high fixed costs, and weaker digital ecosystems can make sophisticated FPE models difficult to implement and sustain. In such cases, policymakers may need to prioritize simpler use cases or shared and cloud-based infrastructure that permits scale economies while preserving operational resilience and broad access (World Bank, 2022). In short, effective FPE strategy depends on matching system design to institutional capacity and market realities.

VII. Conclusion

This paper has developed a comparative typology of fast payment ecosystems built around three institutional models: a unified public model in Brazil, a hybrid multi-infrastructure model in the United States, and a layered infrastructure/platform-led model in China. The central finding is that FPE outcomes depend less on whether provision is public or private than on how governance, interoperability, pricing, and access are aligned to support scale, usability, and trust. Technical capability alone is therefore insufficient; what matters is whether the broader institutional configuration allows network effects to accumulate without generating excessive fragmentation or concentration.

The typology highlights three distinct pathways to scale, each with different strengths and risks. The unified public rail model can support rapid system-wide adoption and formal interoperability when participation, pricing, and standards are aligned from the outset, but it also concentrates operational and fraud-management responsibilities. The hybrid multi-rail model may support resilience, provider diversity, and innovation, while retail diffusion may be slower and less uniform where coordination across rails and front-end platforms is complete and established payment alternatives remain attractive. The layered infrastructure/platform-led model can achieve very rapid retail scale through ecosystem integration and low-friction merchant acceptance, but often with stronger concentration, data-governance, and oversight challenges. These tradeoffs are central to how FPS models should be evaluated, especially by countries considering different institutional pathways for payment modernization. Adoption differences must be interpreted in light of countries' starting points, not as a ranking of underlying payment-system performance or efficiency.

The comparison suggests that assessments of fast payment ecosystem models should take account of domestic institutional capacity, market structure, and policy objectives. The relevant question is not which country model to copy, but how different combinations of governance, interoperability, access, incentives, and oversight perform under local conditions. The paper's typology is intended not as a ranking of models, a source of causal estimates, or a statement of institutional policy advice, but as a comparative framework for examining institutional fit and the tradeoffs associated with different FPE strategies. The findings should therefore be interpreted as conditional comparative considerations rather than recommendations applicable uniformly across countries.

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Fast Payment Ecosystem Models and Performance in Brazil, the United States, and China



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